MODULE 2 · LECTURE 5

CS 6530 Applied Cryptography

Block Ciphers and AES

Module 2 – Symmetric Cryptography, Hashes and Authentication

Block Ciphers and AES

Today's Questions

1. **Why are block ciphers called “block” ciphers?**

   Block ciphers operate on fixed-size blocks of data.

2. **Why did AES replace DES?**

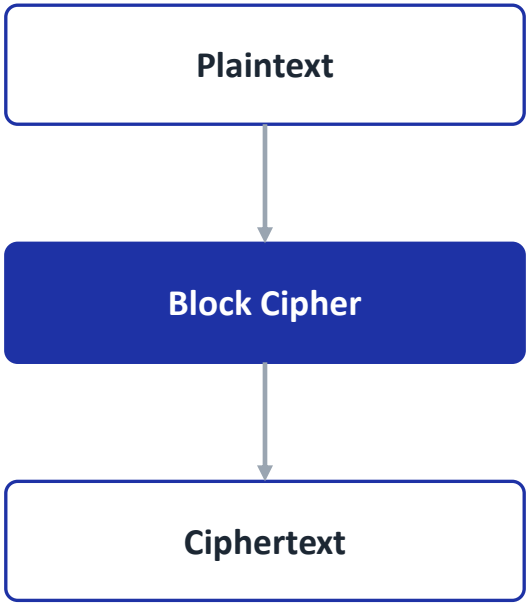
   Engineering evolution from DES to AES.

3. **How does AES transform plaintext into ciphertext?**

   AES architecture: round transformations and key schedule.

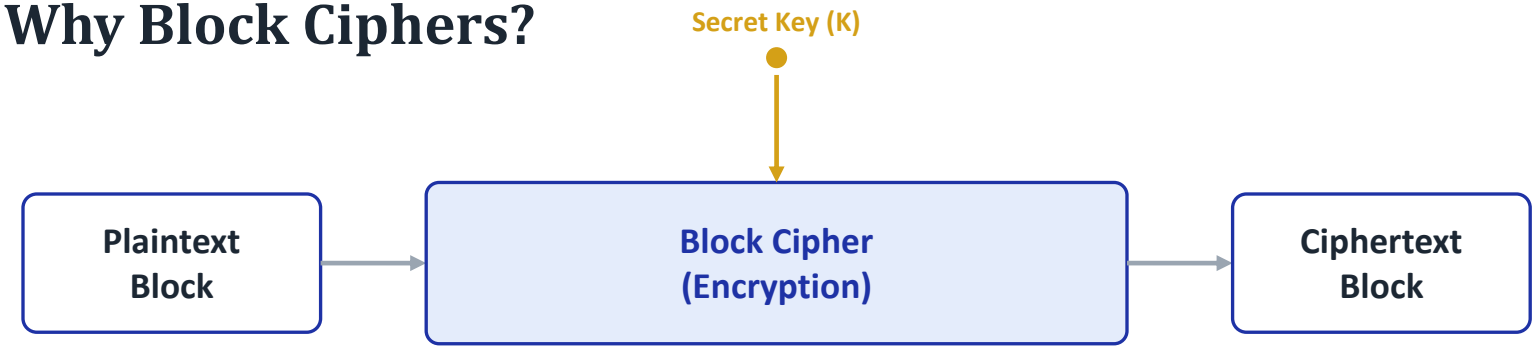
4. **What makes AES secure?**

   Design principles, key sizes and rounds vs. security and performance.

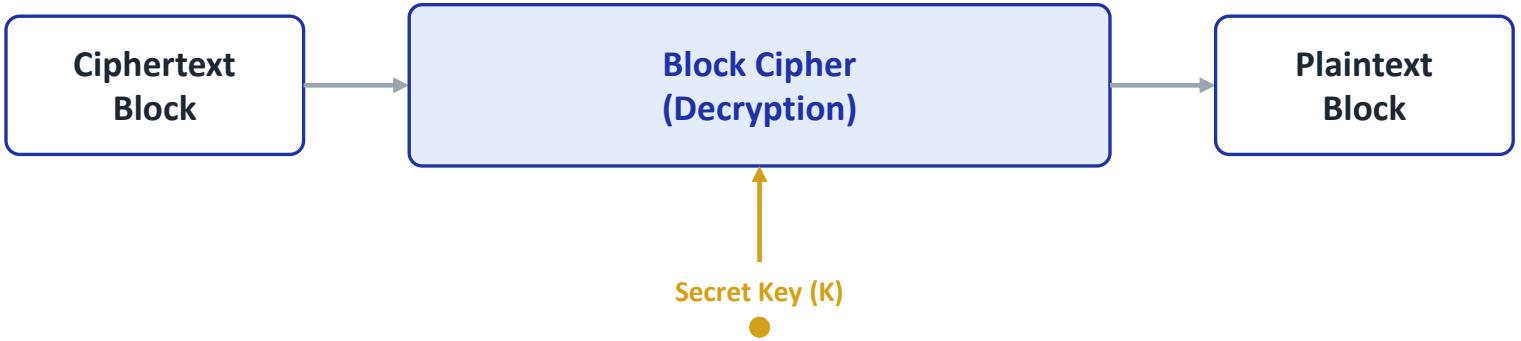


Today's lecture explains how modern block ciphers are designed and why AES became the world's most widely used symmetric encryption standard.

Why Block Ciphers?

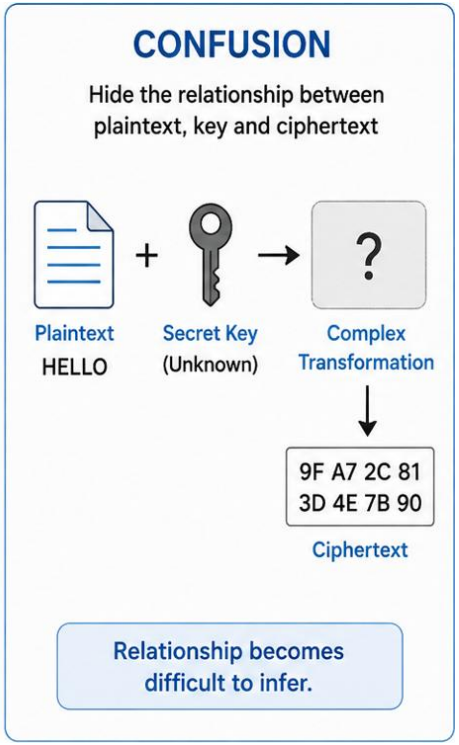


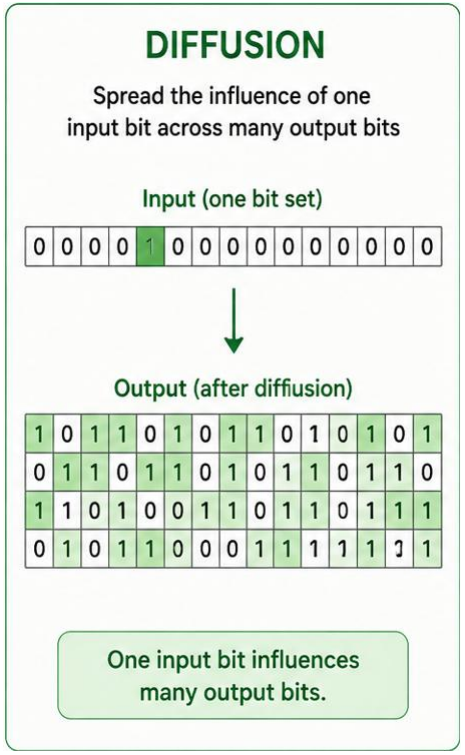
Can we recover the original plaintext?

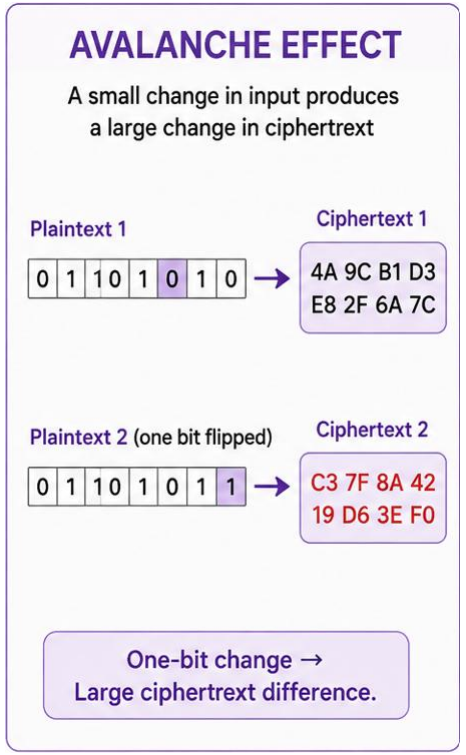


Block ciphers securely transform fixed-size plaintext blocks into ciphertext blocks using a secret key, and the original plaintext can be recovered through decryption.

Principles of Modern Block Cipher Design



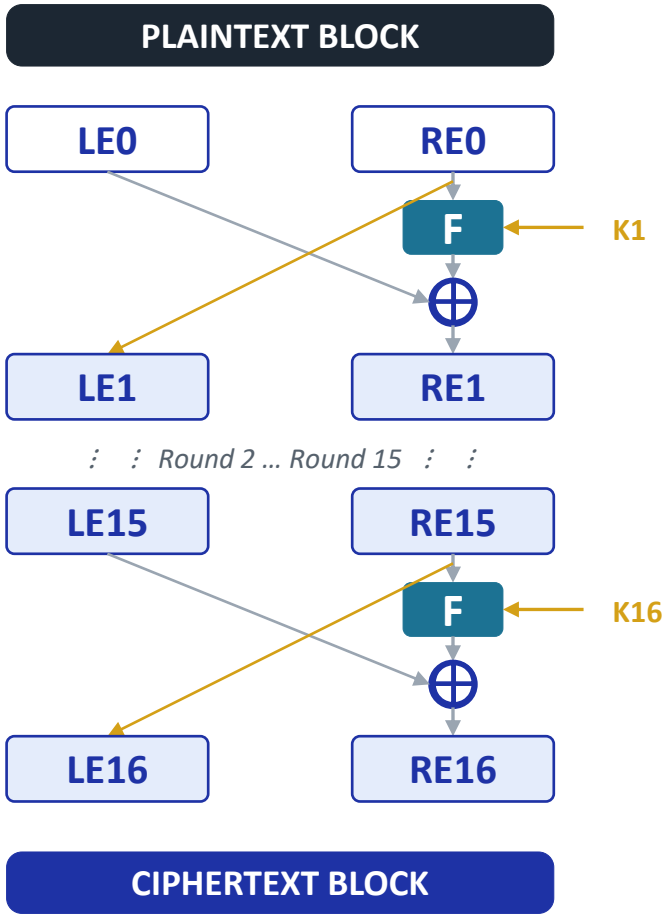




Secure block ciphers are engineered to achieve all three properties simultaneously.

Confusion, diffusion and the avalanche effect are the engineering principles that enable modern block ciphers to provide strong security.

Feistel Network



Engineering Insight

Invertible Function

A function whose original input can be uniquely recovered from its output.

Reversible Process

A process from which the original plaintext can be recovered.

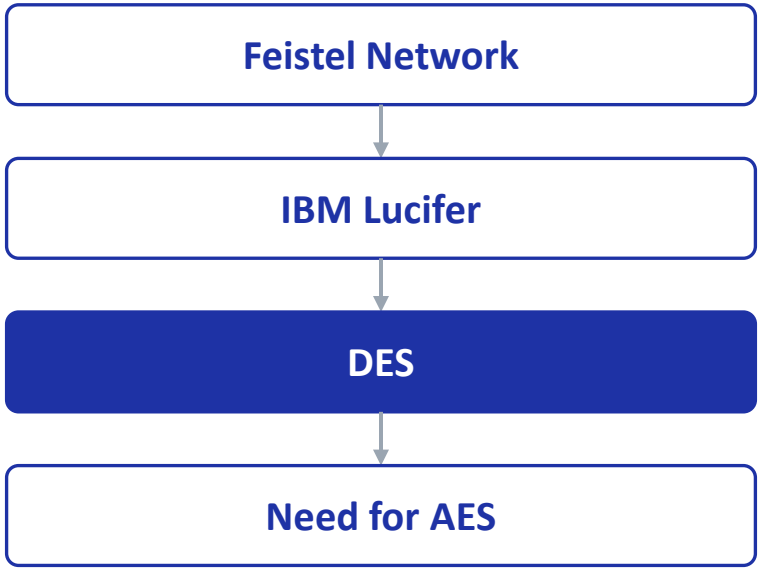
The Round Function need not be invertible, yet the overall Feistel Network remains reversible.

A Feistel Network constructs a secure and reversible block cipher even though its Round Function itself does not need to be invertible.

EVOLUTION OF MODERN BLOCK CIPHERS

DES – The First Widely Adopted Block Cipher

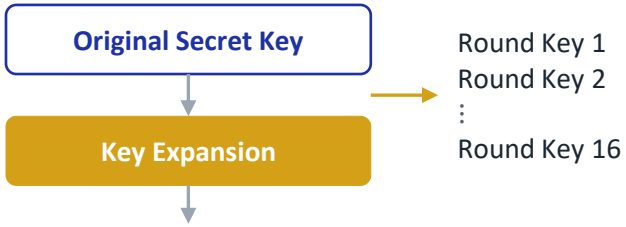
Engineering Evolution



DES at a Glance

Architecture	Feistel Network
Block Size	64 bits
Key Size	56 bits (effective)
Rounds	16

Key Expansion

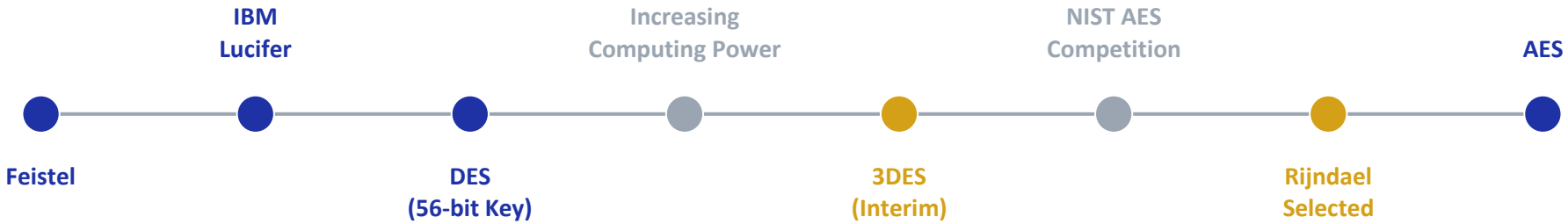




DES demonstrated that the Feistel architecture could provide practical large-scale encryption using multiple rounds and round keys. Advances in computing later motivated the development of AES.

EVOLUTION OF MODERN BLOCK CIPHERS

From DES to AES: Engineering Evolution



Engineering Lesson: Good engineering designs are often extended before they are replaced.

DES	3DES	AES
Original Standard	*Interim Extension*	*New Standard*

Cryptographic standards evolve with technology. DES was extended through 3DES before a completely new standard—AES—was developed through an open international competition.

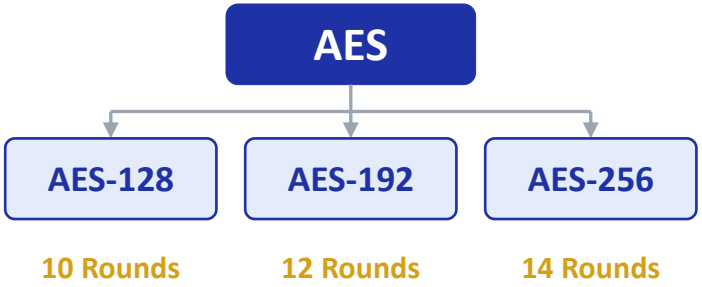
MODERN BLOCK CIPHER STANDARD

AES at a Glance

AES Overview

Standard	FIPS 197
Based On	Rijndael
Designers	Joan Daemen & Vincent Rijmen
Block Size	128 bits
Key Sizes	128 / 192 / 256 bits
Rounds	10 / 12 / 14
Structure	Substitution–Permutation Network (SPN)

AES Family



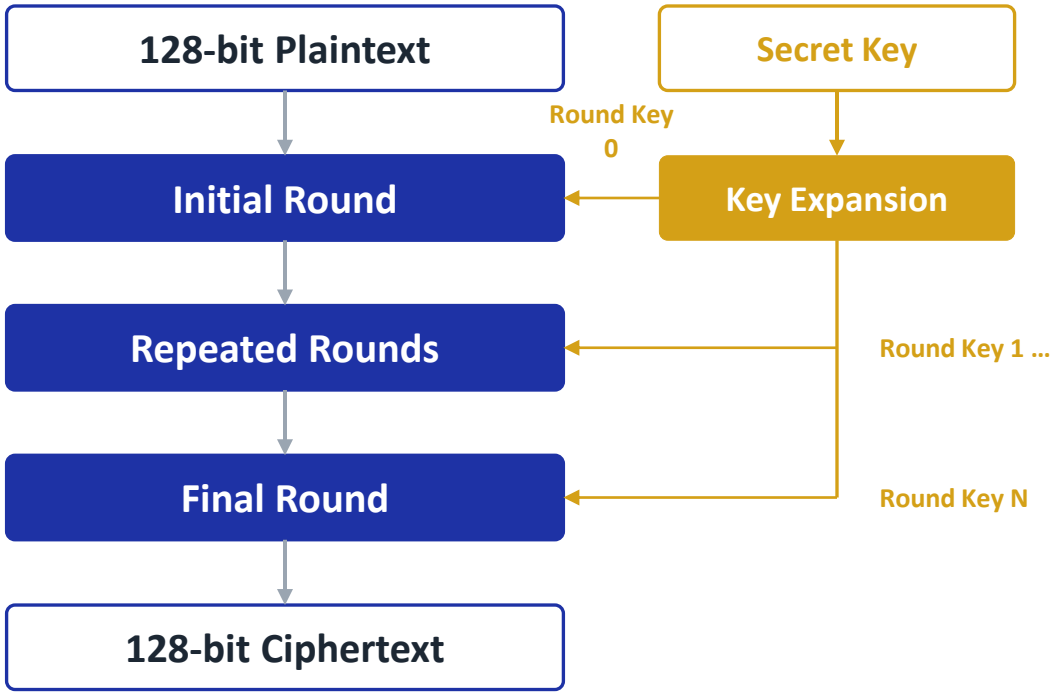
(All use 128-bit blocks)

Engineering Insight

Larger key sizes require additional rounds to provide an increased security margin.

AES is a family of block ciphers that uses a fixed 128-bit block size, selectable key sizes (128, 192 and 256 bits), corresponding numbers of rounds (10, 12 and 14), and a Substitution–Permutation Network architecture.

AES Encryption Architecture



Engineering Insight

AES security is achieved through a sequence of repeated rounds, each using a different round key generated from the original secret key.

AES encrypts a 128-bit plaintext block by combining key expansion with an initial round, repeated encryption rounds, and a final round to produce a 128-bit ciphertext block.

AES ROUND TRANSFORMATIONS

Inside an AES Round

Encryption operates on a single internal AES State, transformed by four operations each round:

1. SubBytes (Byte Substitution)

Before

00	11	22	33
44	55	66	77
88	99	AA	BB
CC	DD	EE	FF

→ S-box →

After

63	82	93	C3
1B	FC	B0	85
97	2B	3D	72
4B	16	28	1E

Each byte is substituted using an S-box.

Purpose: Confusion

2. ShiftRows (Row Shift)

Before

00	11	22	33
44	55	66	77
88	99	AA	BB
CC	DD	EE	FF

No shift →
Shift left 1 →
Shift left 2 →
Shift left 3 →

After

00	11	22	33
55	66	77	44
AA	BB	88	99
FF	CC	DD	EE

Rows are cyclically shifted by different offsets.

Purpose: Rearrangement

3. MixColumns (Column Mixing)

Before

00
55
AA
FF

→ Mixing Matrix $GF(2^8)$ →

After

7A
2C
91
E3

Each column is mixed using a fixed matrix.

Purpose: Diffusion

4. AddRoundKey (Key Addition)

State

7A	2C	91	E3
46	7F	3D	19
C0	5A	E6	6B
B7	2E	8F	4C

⊕ Round Key

3C	A5	7E	2B
9F	2D	C1	6A
5B	7A	8E	15
E2	4F	B7	88

= Updated State

46	89	EF	C8
D9	52	FC	73
9B	20	68	7E
55	61	38	C4

State combined with the round key using XOR.

Purpose: Key Integration

SubBytes
Confusion

ShiftRows
Rearrangement

MixColumns
Diffusion

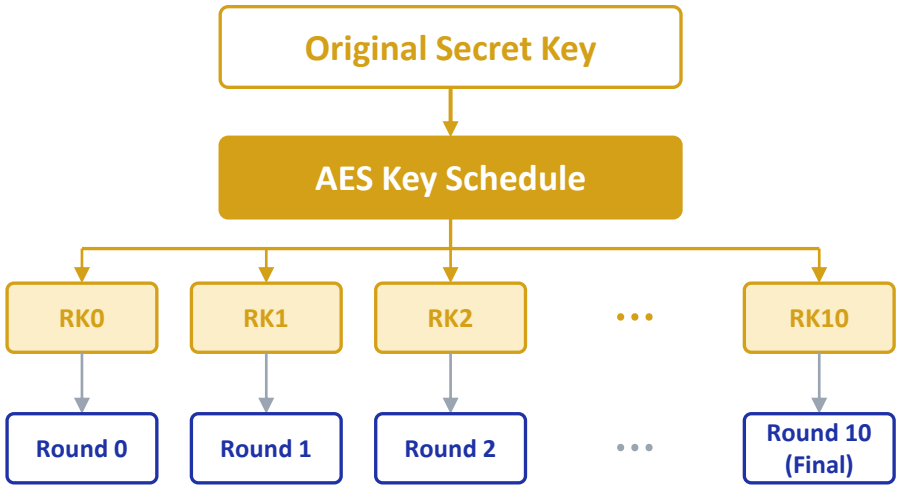
AddRoundKey
Key Integration

Final Round Exception: *MixColumns is omitted in the final round.*

A typical AES round consists of four complementary transformations—SubBytes, ShiftRows, MixColumns, and AddRoundKey. In the final encryption round, the MixColumns transformation is omitted.

AES KEY MANAGEMENT

AES Key Schedule



AES Configuration

AES-128	10 Rounds	11 Round Keys
AES-192	12 Rounds	13 Round Keys
AES-256	14 Rounds	15 Round Keys

Engineering Insight

The user supplies only one secret key. The AES Key Schedule automatically derives all round keys required for encryption.

AES requires only one user-supplied secret key. The Key Schedule automatically generates a unique round key for each encryption round, with larger AES configurations requiring more round keys.

AES Encryption Flow

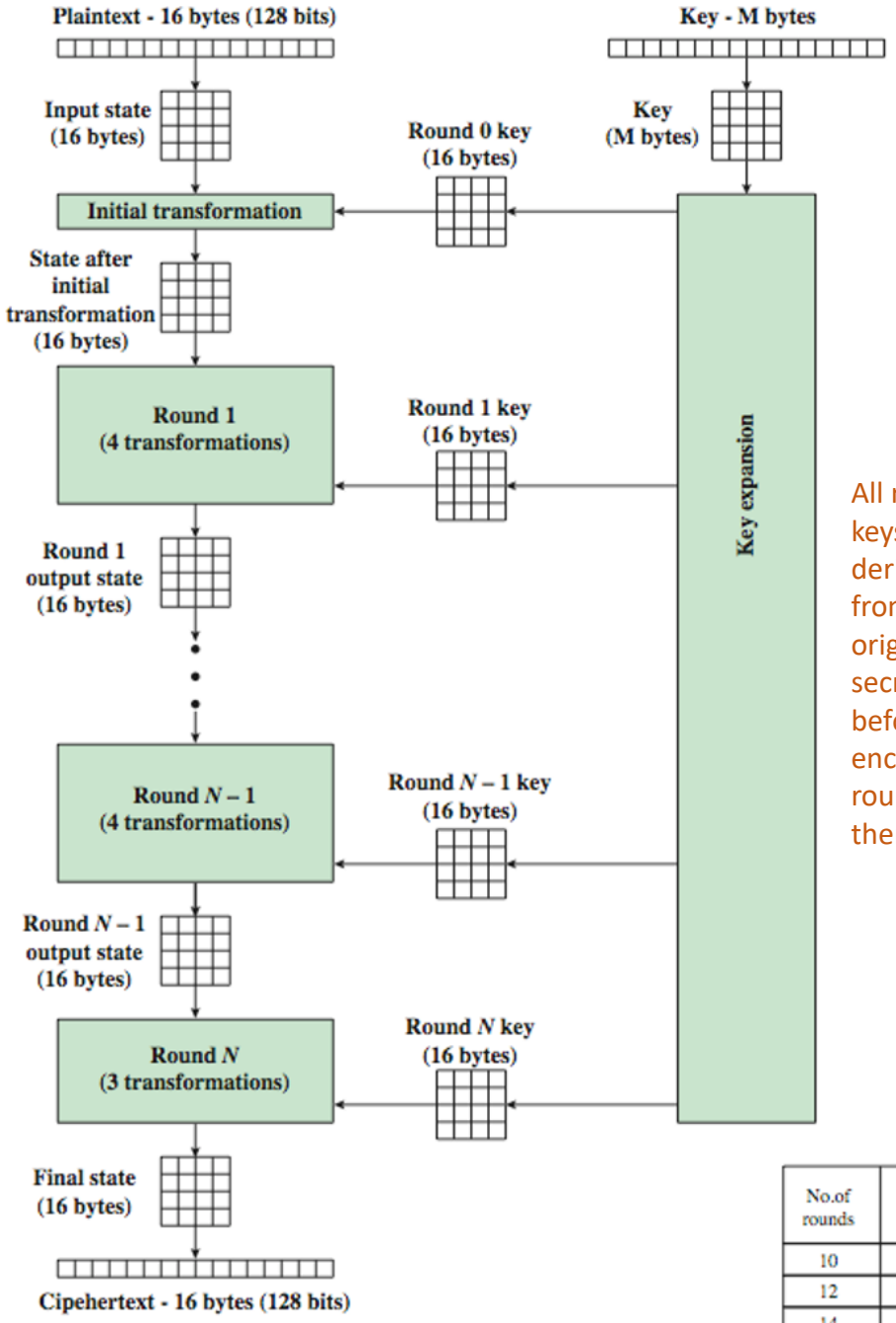
Engineering Insight

AES encrypts data through repeated rounds, with each round using a unique round key derived from the original secret key.

The AES Key Schedule generates a unique round key for every stage of encryption, enabling secure and systematic transformation of the plaintext into ciphertext.

Each encryption round transforms the internal state, while the corresponding round key introduces fresh key material derived from the original secret key.

AES encrypts each block through a sequence of well-defined round transformations, with every stage using a unique round key derived from the original secret key..

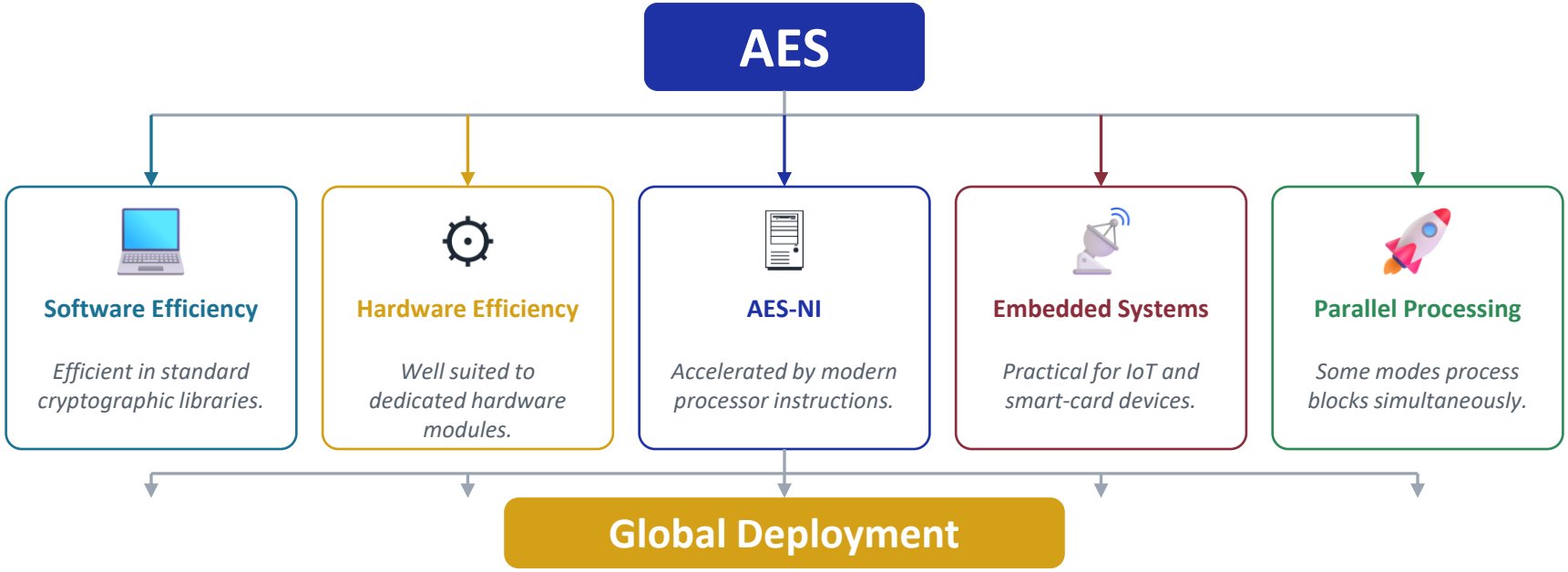


All round keys are derived from the original secret key before encryption rounds use them.

No.of rounds	Key Length (bytes)
10	16
12	24
14	32

ENGINEERING PERSPECTIVE

Why AES Became the Dominant Block Cipher



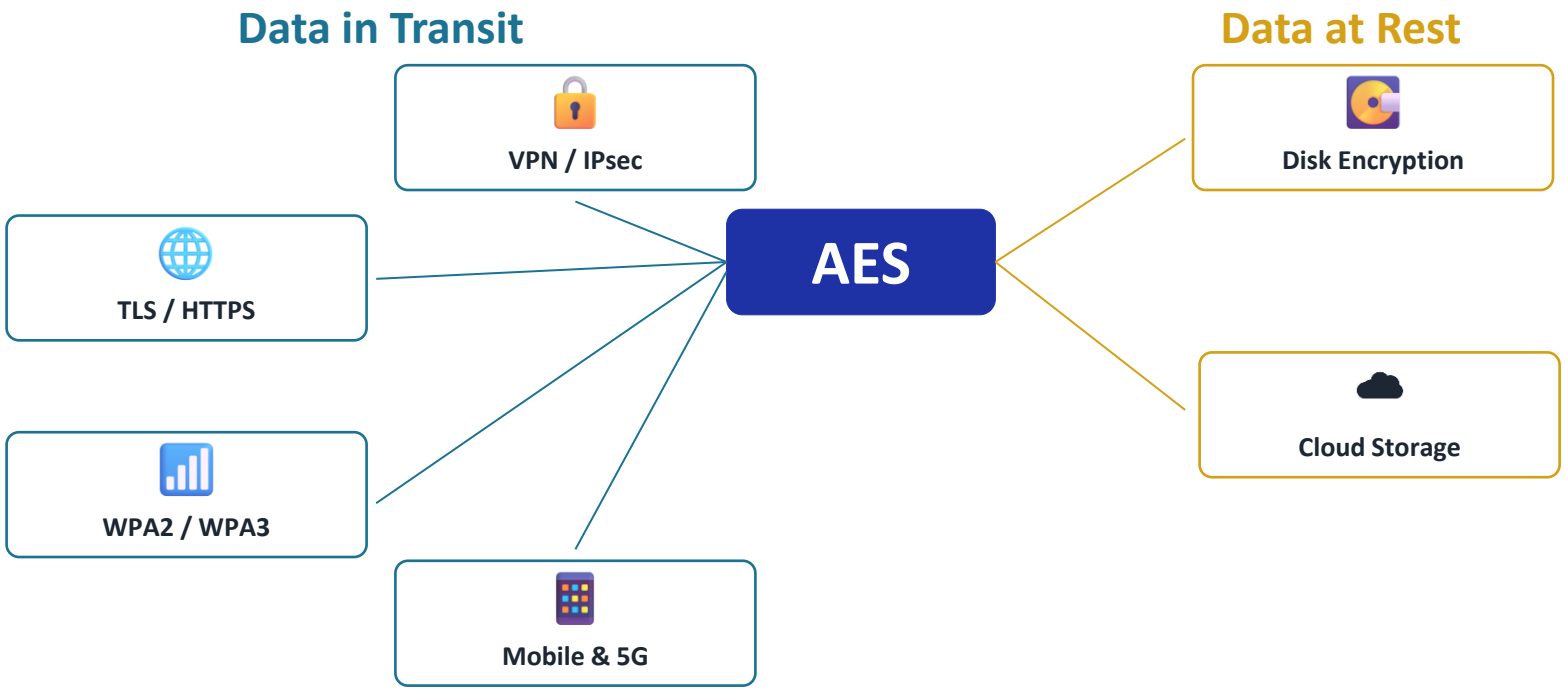
Engineering Insight: AES combines strong security with efficient implementation across diverse computing platforms.

References: FIPS PUB 197; Intel AES-NI Documentation (conceptual reference)

AES became the dominant modern block cipher because it combines strong security with efficient implementation in software, hardware, embedded systems, and modern processors.

APPLIED CRYPTOGRAPHY

AES in Practice



Engineering Insight

AES is rarely used alone — it is the encryption engine within larger secure protocols.

You likely use AES today — browsing, Wi-Fi, cloud, or an encrypted device.

AES is one of the world's most widely deployed cryptographic primitives, protecting data across web communication, VPNs, wireless networks, storage systems, cloud platforms, and mobile environments.

LIVE DEMONSTRATION

AES in Action — Nayuki Excel Demonstration

Nayuki AES Step-By-Step Workbook (illustrative)

00	11	22	33
44	55	66	77
88	99	AA	BB
CC	DD	EE	FF

1. State Matrix
2. SubBytes
3. ShiftRows
4. MixColumns
5. AddRoundKey
6. Ciphertext Output

Demonstration Roadmap

- ✓ Observe the AES State Matrix
- ✓ Follow one encryption round
- ✓ Watch each transformation
- ✓ Observe the Final Round
- ✓ Modify one plaintext byte
- ✓ Observe the Avalanche Effect

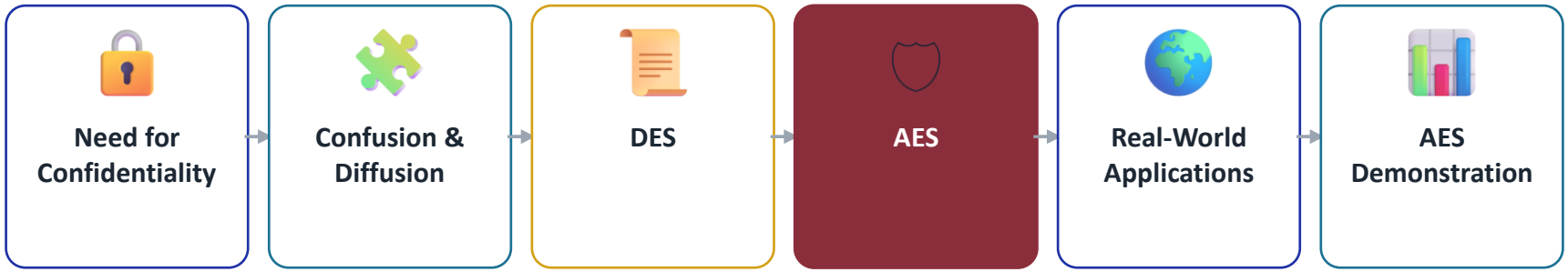
Engineering Note

This demonstration visualizes AES execution to reinforce concepts, not to explain spreadsheet formulas.

The Nayuki workbook transforms the AES concepts from static diagrams into a live execution, helping us visualize how each transformation contributes to secure encryption and demonstrating the avalanche effect.

LECTURE SUMMARY

From DES to AES: The Evolution of Modern Block Ciphers



Engineering Reflection

AES secures one block. Secure systems must securely protect many blocks.

Next Lecture: Modes of Operation

How do we securely encrypt files, packets, and data streams using AES?

Modern block ciphers evolved from strong cryptographic principles into practical engineering solutions. AES is the foundation; the next step is learning how to apply it securely in real communication systems.